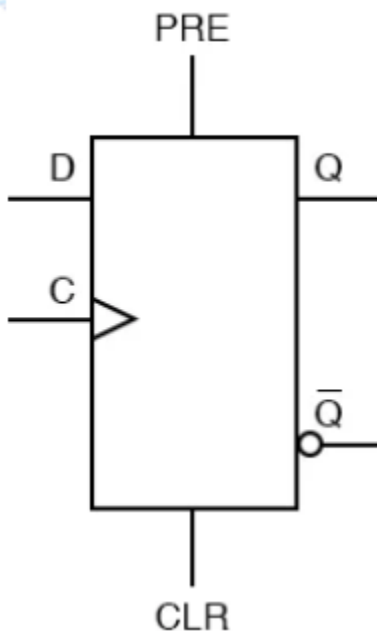


Circuit Simulation Lab:16

Design of Synchronous Circuits

The purpose of this experiment is to introduce the design of Synchronous sequential circuits, in this case Syn-D Flipflop. Synchronous inputs on a flip-flop have control over the outputs (Q and not-Q) only when the clock input has a change either rising edge or falling edge. These inputs are called the preset (PRE) and clear (CLR). The preset input drives the flip-flop to a set state while the clear input drives it to a reset state.



```
module sync_dff(  
input pre , clr , clk , d ,  
output q );
```

```
reg q1 ;
```

```
always @(posedge clk )begin  
if ( clr )  
q1 <= 0 ;  
else if (pre )  
q1 <= 1 ;  
else  
q1 <= d ;  
end
```

```

assign q = q1 ;

endmodule

Testbench
module tb_sync_dff;

reg clk;
reg d;
reg pre;
reg clr;

wire q;

sync_dff dut(
.clk(clk),
.d(d),
.pre(pre),
.clr(clr),
.q(q)
);

always #10 clk = ~clk;

initial begin

$dumpfile("sync_dff.vcd");
$dumpvars(0,tb_sync_dff);

clk = 0;
d  = 0;
pre = 0;
clr = 0;

// Normal DFF Operation
#15 d = 1;
#20 d = 0;

// Asynchronous Preset
#5  pre = 1;
#10 pre = 0;

// Normal Operation
#15 d = 1;

```

```

// Asynchronous Clear
#5 clr = 1;
#10 clr = 0;

// Normal Operation
#20 d = 0;
#20 d = 1;

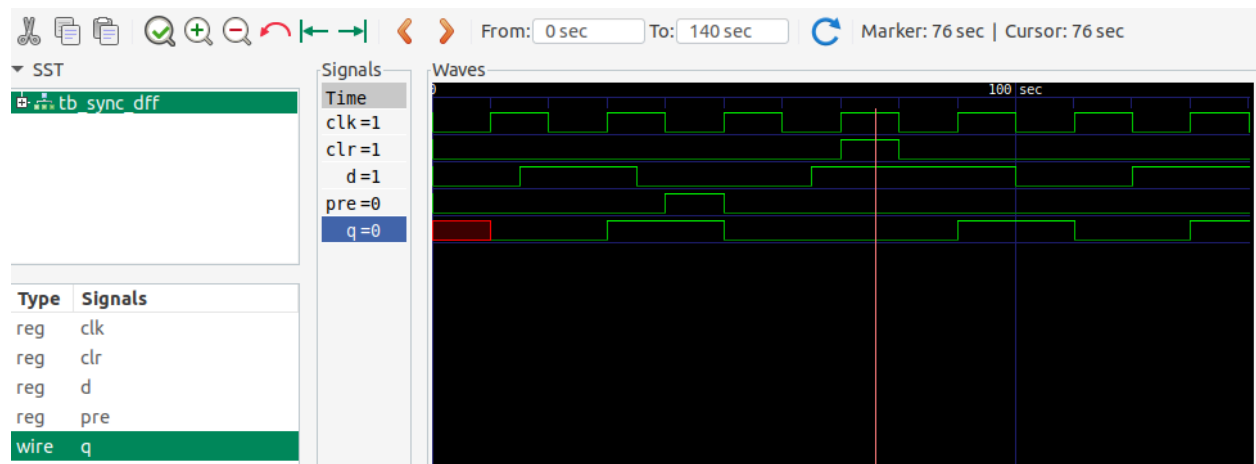
#20;
$finish;

end

initial begin
$monitor("time=%0t clk=%b d=%b pre=%b clr=%b q=%b",
        $time,clk,d,pre,clr,q);
end

Endmodule

```



```
vlsi_dev_box@vlsi-box:~/DOCUMENTS/NIELIT_ASSIGNMENTS/EXP_NO_16$ iverilog -o out
sync_dff.v tb_sync_dff.v
vlsi_dev_box@vlsi-box:~/DOCUMENTS/NIELIT_ASSIGNMENTS/EXP_NO_16$ vvp out
VCD info: dumpfile sync_dff.vcd opened for output.
time=0  clk=0  d=0  pre=0  clr=0  q=x
time=10  clk=1  d=0  pre=0  clr=0  q=0
time=15  clk=1  d=1  pre=0  clr=0  q=0
time=20  clk=0  d=1  pre=0  clr=0  q=0
time=30  clk=1  d=1  pre=0  clr=0  q=1
time=35  clk=1  d=0  pre=0  clr=0  q=1
time=40  clk=0  d=0  pre=1  clr=0  q=1
time=50  clk=1  d=0  pre=0  clr=0  q=0
time=60  clk=0  d=0  pre=0  clr=0  q=0
time=65  clk=0  d=1  pre=0  clr=0  q=0
time=70  clk=1  d=1  pre=0  clr=1  q=0
time=80  clk=0  d=1  pre=0  clr=0  q=0
time=90  clk=1  d=1  pre=0  clr=0  q=1
time=100 clk=0  d=0  pre=0  clr=0  q=1
time=110 clk=1  d=0  pre=0  clr=0  q=0
time=120 clk=0  d=1  pre=0  clr=0  q=0
time=130 clk=1  d=1  pre=0  clr=0  q=1
time=140 clk=0  d=1  pre=0  clr=0  q=1
```